GUEST EDITORS' INTRODUCTION

Special Issue on Interconnects for Chiplet Integration Technologies

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For decades, we have experienced a virtuous cycle of innovation between computing power enabling applications with new or enhanced user experiences. The demand for increased computing capabilities from these applications is relentless, spanning compute power, memory bandwidth, memory capacity, storage, and interconnect bandwidth. Some applications, like data analytics and distributed databases, require performance improvements that double every two to three years, while others, such as supercomputing and artificial intelligence (AI), demand annual exponential progress. For instance, the size of AI language models has exploded from around 1 billion parameters in GPT-2 (2019) to approximately 10 trillion parameters in GPT-4 (2023), a 10,000 × growth in just four years! This virtuous cycle of innovation is expected to continue, with both linear and exponential scaling needed across existing and emerging applications, powered by heterogeneous integration and system optimization.

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Advanced packaging technology is increasingly playing a crucial role in this ongoing progress, along with process technology advancements, enabling the integration of multiple chiplets into commercial products to meet the evolving computing demands within stringent power constraints. Mainstream processors like Intel's 4th Gen Xeon and AMD's EPYC/Ryzen CPUs, along with accelerators and GPUs like AMD Instinct™ MI series, Intel® Data Center GPU Max series, and Nvidia's Hopper series, use chiplets to deliver the desired power-efficient performance. On-package chiplets address challenges, such as die size (reticle) limits, and yield challenges in advanced process nodes. They also facilitate the integration of diverse functions—such as compute and memory—on the same package, optimizing performance and cost-efficiency, and providing bespoke solutions for different market segments.

This special issue on chiplets explores recent advances in the field. It features eight articles, six from industry experts. We believe chiplets-on-package will become the next major platform for innovation, with a universally accepted open standard being crucial to its success. Open standards enable companies to innovate on core technologies without concerns about interoperability with open attach points. Standards like PCI-Express, USB, and Compute Express Link have already transformed the industry at the board level. The industry has embraced Universal Chiplet Interconnect Express (UCIe) as the open standard for chiplet interconnect. Notably, three leading IP providers share their UCIe implementations, showcasing silicon interoperability at events, such as the 2023 Intel Innovation and 2024 Chiplet Summit.

CHIPLET-BASED DESIGN OVERVIEW

The first article, by Phelps and Khan,[A1] provides an overview of chiplets, highlighting their role in meeting AI-driven compute demands. It covers the benefits and challenges of die disaggregation, including economic, power-performance, and time-to-market factors. The article also details Cadence's UCIe IP implementation

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APPENDIX: RELATED ARTICLES

A1. B. Phelps and A. Khan, "Disaggregated designs: Technology challenges and enablers," *IEEE Micro*, vol. 45, no. 1, pp. 9–15, Jan. /Feb. 2025, doi: 10.1109/MM.2024.3524130.

A2. P. Onufryk and S. Choudhary, "UCIe: Standard for an open chiplet ecosystem," *IEEE Micro*, vol. 45, no. 1, pp. 16–25, Jan. /Feb. 2025, doi: 10.1109/MM.2024.3451532.

A3. A. Huynh, K. Stahn, M. Mota, C. de Verteuil, J. Pyon, and R. Movahedinia, "UCIe standard: Enhancing die-to-die connectivity in modern packaging," *IEEE Micro*, vol. 45, no. 1, pp. 26–34, Jan. /Feb. 2025, doi: 10.1109/MM.2025.3526048.

A4. T. C. Carusone et al., "Co-design of interchiplet, package, and system interconnect protocols," *IEEE Micro*, vol. 45, no. 1, pp. 35–40, Jan. /Feb. 2025, doi: 10.1109/MM.2024.3447711.

A5. W. Tang, C. Liu, and Z. Zhang, "Energy-efficient parallel interconnects for chiplet integration," *IEEE Micro*, vol. 45, no. 1, pp. 41–47, Jan. /Feb. 2025, doi: 10.1109/MM.2024.3450841.

A6. D. Jarrett-Amor and T. C. Carusone, "A comparison of single-ended, NRZ unidirectional signaling and single-ended, NRZ simultaneous-bidirectional signaling for die-to-die links," *IEEE Micro*, vol. 45, no. 1, pp. 48–56, Jan. /Feb. 2025, doi: 10.1109/MM.2024.3436008.

A7. A. Smith et al., "Interconnect design for heterogeneous integration of chiplets in the AMD instinct MI300X accelerator," *IEEE Micro*, vol. 45, no. 1, pp. 57–66, Jan. /Feb. 2025, doi: 10.1109/MM.2024.3462351.

A8. S. Muthrasanallur and Y. Zorian, "A test, debug, and silicon lifecycle management architecture for a UCIe-based open chiplet ecosystem," *IEEE Micro*, vol. 45, no. 1, pp. 67–74, Jan. /Feb. 2025, doi: 10.1109/MM.2024.3435702.

across process nodes, addressing interoperability and validation in presilicon and silicon.

CHIPLET INTERFACE

The next three articles explore UCIe as a chiplet interface. Onufryk and Choudhary,[A2] from Intel, provide an in-depth look at UCIe specification, covering physical layer, channel, link layer reliability, protocol mapping, usage models, form factors, and compliance for both planar and vertical chiplets. They also discuss the system-level integration challenges around manageability and debug that UCIe addresses. They envision composable systems at the rack and pod level, enabled by copackaged optics with UCIe-based retimers. Huynh et al.,[A3] from Synopsys, present their UCIe implementation, design flow, bump map, package routing, and silicon measurements. Carusone et al.,[A4] from Alphawave, focus on UCIe IP implementation, detailing design flows and solutions for signal integrity at both silicon and package levels.

ENERGY-EFFICIENT CHIPLET INTERCONNECT

The next two articles from academia focus on improving energy efficiency, a key metric for chiplet adoption and proliferation. Tang et al.,[A5] from the University of Michigan, provide circuit architecture details along with their silicon learnings of the AIB standards and discuss how to migrate those to UCIe with a power target of 0.1 pJ/b for planar interconnect. Jarrett-Amor and Carusone,[A6] from the University of Toronto, conduct a comparison between different signaling methods for driving better power efficiency, as the data rates will increase for planar implementations in the future.

WE BELIEVE CHIPLETS-ON-PACKAGE WILL BECOME THE NEXT MAJOR PLATFORM FOR INNOVATION, WITH A UNIVERSALLY ACCEPTED OPEN STANDARD BEING CRUCIAL TO ITS SUCCESS.

CHIPLET-INTEGRATED PROCESSOR DESIGN

Smith et al.,[A7] from AMD, offer a comprehensive overview of the AMD Instinct™ MI300X accelerator, including insights on manufacturing, architecture, circuit design, and silicon measurements from real workloads. They emphasize the importance of heterogeneous integration for emerging AI/machine learning usage models and explain the rationale behind various integration techniques, such as 3-D logic-on-logic,

high-bandwidth memory on 3-D logic, and planar logic-to-logic connectivity.

CHIPLET-BASED DESIGN ECOSYSTEM

In the last article, Muthrasanallur and Zorian,[A8] from Intel and Synopsys, respectively, explore the challenges in testing, debugging, and managing the entire silicon lifecycle, from sort to field, for a thriving chiplet ecosystem. They detail how UCIe ensures compatibility across multiple standards (e.g., PCIe, USB, JTAG) and IP blocks with a unified software and hardware interface, promoting a secure, cost-effective, and scalable plug-and-play ecosystem.

This special issue aims to provide a comprehensive overview of the challenges and opportunities in the open chiplet ecosystem. Chiplets are transforming system-on-chip construction, with packaging emerging as the platform for future innovation. Driven by open standards, chiplet-based ecosystems promise multidecade advancements. We hope these articles inspire further research in this evolving field. We would like to thank all of the authors who submitted articles to this special issue, the reviewers for their valuable insights and feedback, the Editor-in-Chief (Hsien-Hsin Sean Lee and Lizy Kurian John before him) and the *IEEE Micro* staff for their support in making this special issue happen.

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